

Flexible framework

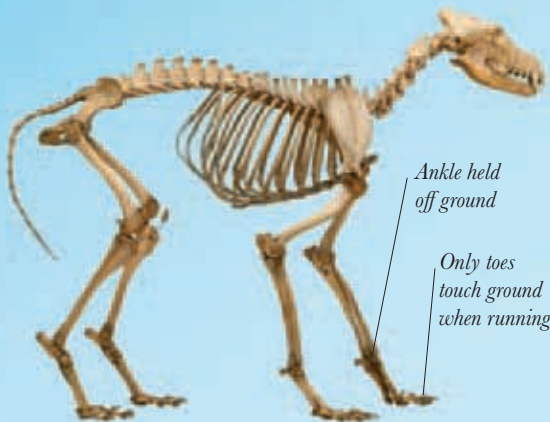
A harbor seal's skeleton has more than 200 different bones, which meet at junctions called joints. Some bones—particularly in the skull—are fused together, so that the joints are fixed and cannot move. Other bones are tipped with slippery cartilage where they join, which enables the bones to move smoothly against one another. Altogether, the skeleton makes up less than one-tenth of the seal's total weight.



Front flipper with five fingers

Long-legged runner

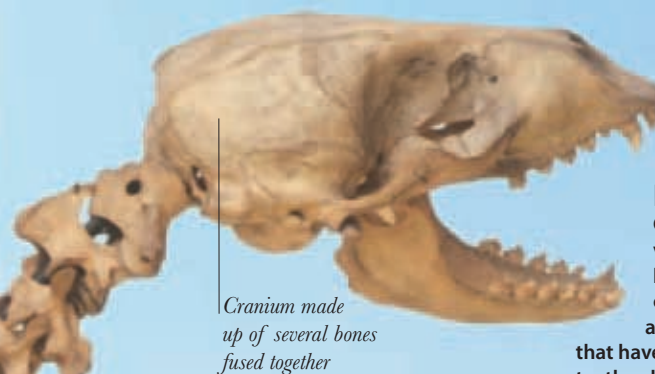
This wolf skeleton has the same limb bones as those of a harbor seal, but their proportions are very different. The wolf's main leg bones are much longer than the seal's, while its toes are shorter—two adaptations that make the wolf a fast and tireless runner.



Ankle held off ground

Only toes touch ground when running

Skulls



Cranium made up of several bones fused together

Big brain case

Compared to most other vertebrates, mammals have a large brain case, or cranium. Mammals are the only vertebrates that have differently shaped teeth, which they use for cutting, gripping, and grinding up food.



Parrot's double-hinged beak opens wide for feeding

Hinged beak

Birds' beaks are made of bone covered by keratin—the same substance found in feathers and claws. In some birds, such as parrots, both halves of the beak hinge against the skull, allowing the birds to open their mouths wide when feeding.

Large jaws

A crocodile's jaw is made of several bones fused together. Unlike mammals, this reptile has conical teeth that it continually sheds and replaces during its lifetime.

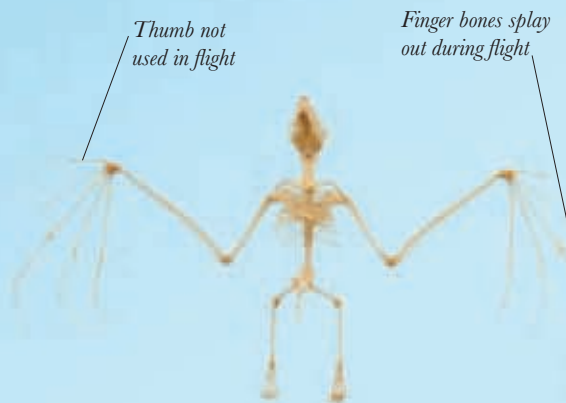


Anchorage for jaw muscles

Crocodile's conical teeth tear up meat

Lightweight skeleton

Animals with bony skeletons all have gristle, or cartilage, in their joints. Sharks, skates, and rays, however, are different. Although they are vertebrates, their entire skeletons are made of cartilage, instead of bone. This kind of cartilaginous skeleton is not as strong as a bony one, but it is much lighter. Even so, open-water sharks often swim continuously, partly to keep themselves afloat.

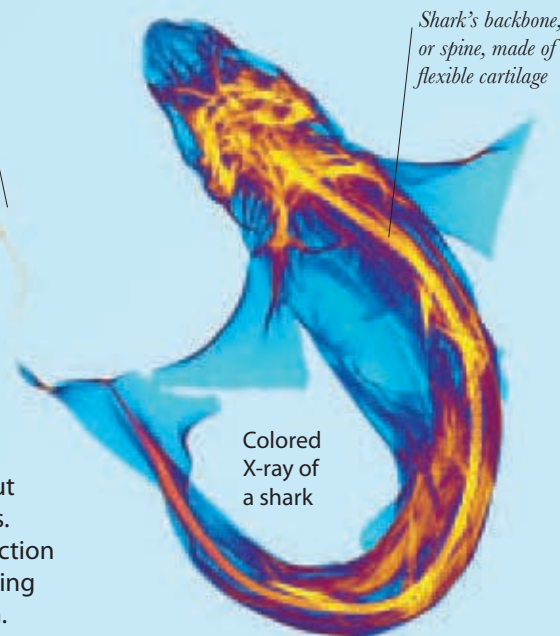


Thumb not used in flight

Finger bones splay out during flight

Flick of the fingers

A bat's wings are made of skin, stretched out by extremely long and slender finger bones. During flight, a bat can quickly change direction by flexing its eight fingers and giving the wing tips an extra flick as they beat up and down.



Shark's backbone, or spine, made of flexible cartilage

Colored X-ray of a shark